

Installation Instructions Downhole & Vertical Cable

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PRODUCT CODE

NTE4**JL096xx¹

NME4**JL048xx¹

APPLICATION RECOMMENDATIONS

This type of cable is designed to have its mass supported through the sheath material coupled to round GRP peripheral strength members to achieve long spans in vertical applications. The range can support its weight up to 2 km of vertical drop. These cables are favoured where an element of mechanical robustness is required due to the application environment.

ENVIRONMENTAL CONSIDERATIONS

If the cable is outdoors, it may be necessary to ensure the sheath is UV resistant, has a termite resistant jacket or in some cases is protected against rodent attack. For termite resistance, a Polyamide (Nylon) PA 12 jacket of appropriate thickness has been proven acceptable for Australian species – sometimes requiring a sacrificial sheath to protect the necessary pristine Nylon surface during the installation process. When employed with GRP armour, the Nylon is positioned under the armouring for this purpose.

The GRP armour serves multiple purposes including rodent resistance and protection against bird attack. It has high-performance mechanical characteristics to withstand harsh environments and used for high tensile applications.

INSTALLATION AND TERMINATION CONSIDERATIONS

The cable can be installed via single point suspension unsupported throughout its length or vertically supported with external lashings or ties. Clipping or strapping the cable to a vertical object can be performed in many ways and there are no hard and fast rules other than the specified mechanical limits for crush/impact/bend/tension, which must not be exceeded.

Special aerial fittings for pole or tower mounting, to reinforce the termination and to mount the cable for this type of application are commercially available. They are proven on horizontal applications - which are generally more severe than vertical drops. For the vertical installations, cable is un-reeled from a drum, over a barrel anchored above the hole, but free to rotate. When the cable length required reaches the bottom, the barrel should be locked to prevent further rotation and the cable shall be wrapped multiple times around the barrel and clamped. The end terminations shall be managed following the guidelines below.

To ensure clean termination enclosures at the lower end of a vertical drop, that is, to prevent any form of gel leakage due to temperature fluctuations or otherwise, there are a few practices that should be considered. These include bringing the cable into the joint enclosure from below, forming the excess cable into a loop at the lower end and/or designing a short horizontal at the end of the drop.



¹ Refer to relevant AFL cable technical data sheet for corresponding Product Codes